

Lab 1: Fundamentals of Object-Oriented Programming

Paweł Głaba, PhD

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Basic Problems

1. Implement a class `Box` with three instance variables: `width`, `height`, and `depth` (all of type `double`), with the default value of `1.0`. Create two constructors: a default constructor (no parameters) and a three-parameter constructor (`width`, `height`, and `depth`).

Implement the following methods:

- `getSurfaceArea()`, which returns the total surface area of the box,
- `getVolume()`, which returns the volume of the box.

Write a test program called `BoxTester` (new class) and demonstrate the functionality of the `Box` class.

2. Implement a class `PrepaidCard`, which models a prepaid card with one instance variable `balance` (integer) with a default value of `100`. Create two constructors: a default constructor and a one-parameter constructor for setting the balance.

Implement the following methods:

- `topUp(amount)`, which increases the balance,
- `pay(amount)`, which decreases the balance; if `amount > balance`, print `"Insufficient funds"`.

Write a test program called `PrepaidCardTester` (new class) and demonstrate the functionality of the `PrepaidCard` class.

Extra points

Add the method `transferTo(anotherCard, amount)` to transfer the given amount from this `PrepaidCard` to another `PrepaidCard`.

Space Station Management

Imagine that you are responsible for managing a futuristic space station. In the following lists, you will gradually expand this program throughout the semester.

0.1 Commander

For now, your task is to complete the first step: a crew profile generator. These profiles will help organize the crew, support daily operations, and prepare the system for future extensions.

The profile should include the following characteristics:

- name,
- age,
- role (e.g. Commander, Pilot, Engineer),
- specialization,
- mission hours.

Hint: In Java, the `role` field can be implemented as an `enum` instead of a plain `String`.

The program should print out a complete profile of the crew member. Example:

Hello! Meet [Name], a [Age]-year-old [Role]. This crew member specializes in [Specialization] and has already completed [MissionHours] mission hours.

0.2 Quartermaster

As a quartermaster, you are responsible for distributing oxygen supplies to crew members. Implement a method that calculates the daily oxygen allocation based on the crew member's mission hours. Use the number of daily distributions as a parameter of the method.

Assume that:

- a crew member needs 2 oxygen units per day by default,
- if the crew member has more than 1000 mission hours, they need 3 oxygen units per day,
- if oxygen is distributed twice a day, the total amount should be divided equally.

Print out the result in the following format:

[Name] needs X oxygen units daily, which means Y oxygen units per distribution.

0.3 Station Tracker – main method

In the `main` method, declare an array for 10 crew members on your space station. Each crew member should correspond to one index in the array. Populate the array with some initial data.

Write a loop that goes through the array and prints the details of each crew member in a readable format.

At the end, print:

- the average age of the crew,
- the average number of mission hours.